Module 09 Lab Exercise - Contribution Journal

John Grubbs

I completed and ran my own copy of the Module 09 Ensemble Methods lab using the Iris dataset. I loaded the data, separated the features and target values, and split the dataset into training and testing sets using a 70/30 split.

I trained and evaluated a single DecisionTreeClassifier. The model reached 100.00% accuracy on the test data. I then trained a RandomForestClassifier using 100 decision trees and compared its performance with the single Decision Tree. The Random Forest also reached 100.00% accuracy.

I extracted the feature importance values from the Random Forest and created a bar chart to display them. Petal width and petal length were the two most important features. I completed the knowledge check by explaining ensemble methods, comparing the two models, and interpreting the feature importance results.

Keira Burdine

I independently completed the assigned tasks in Google Colab by reviewing the instructions, correcting the Python code formatting and trained both the Decision Tree and Random Forest models using the Iris dataset.
I evaluated the accuracy of both models and successfully generated a feature importance graph to compare the different flower characteristics. When I encountered errors, I reviewed the code, identified the issues and made corrections, while also reruning the cells to ensure they worked successfully.
I also completed the Knowledge Check questions based on the results of the lab. My contribution included completing the coding activities, troubleshooting errors, analyzing the results and documenting what I learned throughout the assignment